GG-R-1472B
30 April 1987
SUPERSEDING
GG-R-1472A
16 June 1975

FEDERAL SPECIFICATION

RETAINER, MATRIX, DENTAL (Tofflemire)

This specification is approved by the Commissioner, Federal Supply Service General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers matrix retainers of Tofflemire design suitable for use in dental operative procedures.

1.2 Classification.

1.2.1 Styles and sizes. The retainers covered by this specification shall be of the following styles and sizes as specified (see 6.1):

Style A. Tofflemire, Universal, Straight
Style B. Tofflemire, Contra-Angle

Size 1. Adult
Size 2. Junior

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitations for bids or requests for proposal, form a part of this specification to the extent specified herein.

AMSC N/A FSC 6520

|Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-RST, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.|

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Federal Specifications

GG-B-95	- Band Set, Matrix, Dental, Strip, Matrix Dental, and Band, Matrix, Dental.
GG-I-526	- Instruments, Dental and Surgical, General Specifications.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

Federal Standards

FED. STD. No. 123 - Marking for Shipment (Civil Agencies).
FED. STD. No. 151 - Metal, Test Methods.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards and commercial item descriptions outlined under General Information in the Index of Federal Specifications and Standards and Commercial Item Descriptions. The Index, which includes cumulative bi-monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402).

(Copies of listed federal and military standards, specifications, Commercial Item Descriptions (CIDS), handbooks and associated documents listed in the Department of Defense Index of Specifications and Standards (DoDISS), should be obtained from the DoD Single Stock Point, Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120. Copies of industry association documents should be obtained from the sponsor. Copies of all the listed documents should be obtained from the contracting activity or as directed by the contracting officer).

(Federal Government activities may obtain copies of Federal Standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies).

Military Specifications

MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.
MIL-P-116	- Preservation, Methods of.

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Military Standards

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized Unit Loads.
MIL-STD-794	- Parts and Equipment, Procedures for Packing of.

DRAWINGS

DEFENSE PERSONNEL SUPPORT CENTER (DPSC)

24349	Retainer, Matrix, Dental, Tofflemire, Universal, Straight.
24350	Retainer, Matrix, Dental, Tofflemire, Contra-Angle, Adult.
24351	Retainer, Matrix, Dental, Tofflemire, Contra-Angle, Junior.

(Copies of military specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or requests for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 18	Standard Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.
E 92	Standard Method of Test for Vickers Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 General requirements. The retainers shall be in accordance with applicable requirements of GG-I-526 and as specified herein.

3.2 Drawings. The drawings forming a part of this specification are specification control drawings. The supplier shall be responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in improper design or incorrect fits, the supplier shall provide a retainer with dimensions within the tolerances prescribed on the drawing to insure proper design, function, fit, assembly and operation of the item. No deviation from the prescribed dimensions and tolerances is permissible without prior approval of the contracting officer.

3.3 Material. Retainers shall be fabricated of Class 3, austenitic stainless steel. The frame shall be fabricated from Type 302 stainless steel. The vise, screw and nut shall be fabricated from Type 303 stainless steel.

3.4 Hardness. The hardnesses of the components shall be entirely suitable for the purpose intended except that no components shall have a Rockwell hardness less than B85 when tested as specified in 4.2.4.2.

3.5 Style, design and dimensions. Retainers shall be of the Tofflemire design. The style, design and dimensions of the retainers shall conform to the Defense Personnel Support Center (DPSC) drawings 24349, 24350 and 24351 for the applicable style and size. The retainers shall be suitable for securely holding matrix bands around a tooth during dental operative procedures and for providing support for plastic silver amalgam during compacting required for the formation of a metallic restoration. Retainers shall be suitable for use with bands conforming to GG-B-95, as specified in Table I.

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TABLE I. Drawing numbers and matrix bands for applicable retainer

Retainer	Drawing Numbers	Matrix Bands (GG-B-95)
Style A (Tofflemire, Universal, Straight)	24349	Type II, Bands for adult size Tofflemire matrix retainers
Style B (Tofflemire, Contra-angle) Size 1-Adult	24350	Type II, Bands for adult size Tofflemire matrix retainers
Size 2-Junior	24351	Type II, Bands for junior size Tofflemire matrix retainers

3.6 Finish. The retainers shall have a mirror or mirror and satin finish except for threaded or knurled portions. All surfaces and edges shall be smooth to prevent pinching or scratching of mouth tissue. Knurled portions of nut and screw shall be diamond or straight knurl.

3.7 Corrosion-resistance. Retainers shall comply with the boil test requirements of GG-I-526, when tested as specified in 4.2.4.3.

3.8 Identification marking. The retainers shall be marked in accordance with GG-I-526 and as specified herein. Each retainer shall be legibly and permanently marked with word "Tofflemire" on the frame. Retainers of foreign manufacture shall also be marked "stainless".

3.9 Workmanship. The retainer shall be free from defects which detract from its appearance or may impair its serviceability, proper functioning, or intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein.

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Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract for a period of 3 years after delivery of supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the component or material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the contractor, the contract number, the name of the component or material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables II, III and IV (see 4.2.1, 4.2.2 and 4.2.2.1), and table V (see 4.2.3).

TABLE II. Inspection and acceptable quality levels

Type Defect	Inspection level	AQL (Percent Defective)
Major A	II	1.0
Major B	S-2	4.0
Minor	II	4.0

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4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQLs) shall be as indicated in table II. The unit of product for examination purposes shall be one retainer.

4.2.2 Examination. The retainers shall be examined to determine compliance with specification requirements. Examination shall include, but shall not be restricted to, the defects listed in table III.

TABLE III. Classification of defects

Categories	Defects
Critical:	None defined.
Major A:	
101	Style and design not as specified for style and size of retainer.
102	Components missing, malformed, fractured or otherwise damaged.
103	Threaded portions bind or are not free moving.
104	Threads of threaded components stripped, not properly formed, do not properly engage or threading not free of foreign matter.
105	Sliding components do not slide smoothly.
106	Retainer not free from cracks or burrs.
107	Finish not as specified.
108	Edges and surfaces not smooth to prevent pinching or scratching of mouth tissue.
Major B:	
151	Dimensions of retainers not as specified.
Minor:	
201	Retainer not permanently and legibly marked with name or trademark of the contractor.
202	"Tofflemire" marking not present nor properly located.

4.2.3 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table IV. The unit of product for testing purposes shall be one retainer. The acceptance number shall be zero for all sample sizes. Where feasible, the same sample unit shall be used for determination of two or more test characteristics.

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TABLE IV. Sampling for tests

Characteristic	Requirement	Test Procedure	Inspection level
Material	3.3	4.2.4.1	4.2.3.1
Hardness	3.4	4.2.4.2	S-2
Boil Test	3.7	4.2.4.3	S-2

4.2.3.1 Sampling of raw materials. A sufficient quantity of each lot of metal used in the construction of the retainers shall be selected for chemical analysis.

4.2.4. Tests. Tests shall be conducted to determine compliance with the requirements of this specification (see 4.2.3).

4.2.4.1 Chemical composition. In the absence of certificates of quality for the raw material, chemical analysis shall be conducted in accordance with method 111 or 112 of FED-STD-151. In case of dispute, method 111 shall govern.

4.2.4.2 Hardness. Hardness shall be determined in accordance with ASTM Standard E18 (Rockwell) or E92 (Vickers).

4.2.4.3 Boil test. The retainers shall be subjected to the boil test, as specified in GG-I-526.

4.3 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Each retainer shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Preservative application. Preservatives shall not be used.

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5.1.1.3 Unit package. Each retainer shall be packaged in accordance with MIL-P-116, method IC-3.

5.1.1.4 Intermediate package. Twelve unit packages containing retainers of one style and size only shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.2 Level C.

5.1.2.1 Unit package. Preservation for retainers shall conform to the requirements of MIL-STD-794 for level C.

5.1.2.2 Intermediate package. Twelve unit packages containing retainers of one style and size only shall be packaged in accordance with the requirements of MIL-STD-794 for level C.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. Thirty intermediate packages (360 units) containing retainers of one style and size only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type, PPP-B-621, class 2 or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. Thirty intermediate packages (360 units) containing retainers of one style and size only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type, PPP-B-621 class 1 or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. Thirty intermediate packages (360 units) containing retainers of one style and size only shall be packed in accordance with the requirements of MIL-STD-794 for level C.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

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5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination exceeds 250 lbs (excluding the pallet) or 20 cubic feet. Loads shall be unitized as specified in MIL-STD-147, or Type IV or Type V, 4-way entry pallets. Pallet shall have a length of 40 inches and a width of 48 inches. Pallet loads, including the pallet, shall not exceed 54 inches in height, 40 inches in length and 48 inches in width. Quantity for shipment to one destination of less than 250 lbs or 20 cubic feet need not be palletized.

5.2.5.1 Levels A and B. Retainers packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets as specified in 5.2.5.

5.2.5.2 Levels C. Retainers packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

5.3 Marking

5.3.1 Civil Agencies. Marking shall be as specified in Fed. Std. No. 123.

5.3.2 Military activities.

5.3.2.1 Levels A, B, and C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.3.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Style and size required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 National Stock Number (NSN) coverage. This specification covers the following items listed in the Federal Supply Catalog.

National Stock Number	Item Identification
6520-00-552-3150	RETAINER, MATRIX, DENTAL, Contra-Angle, Tofflemire, Adult Size
6520-00-687-8107	RETAINER, MATRIX, DENTAL, Contra-Angle, Tofflemire, Junior Size
6520-00-985-7123	RETAINER, MATRIX, DENTAL, Universal, Straight, Tofflemire

6.3 Subject term (key word) listing.
Adult Size Retainer
Contra-Angle Retainer
Dental Matrix
Junior Size Retainer
Tofflemire
Universal

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

Review Activities:

Army-MD
Navy-MS
Air Force-03

CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS
PHS
FDA-MPFQAS

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

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